Md Shibly Sadique

PhD Candidate, Biomedical Image Analysis, Old Dominion University

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PhD candidate with eight years of research experience in medical image analysis and machine learning, graduating July 2027. Demonstrated expertise in designing, training, and evaluating deep learning models with PyTorch, TensorFlow, and Scikit-learn for 3D volumetric MRI and CT. Authored fifteen peer-reviewed publications and led a team in an international segmentation challenge. Aiming to apply uncertainty-aware modeling skills to deliver reliable clinical AI systems.

Experience

Old Dominion University - *Graduate Research Assistant, Vision Lab* — *Norfolk, VA* **2018 - Present**

- Developed UNCE-NODE, a neural ODE-enhanced U-Net for multimodal MRI tumor segmentation, cutting trainable parameters by roughly 30% and accelerating inference while holding Dice accuracy.
- Built a classifier separating recurrent brain tumor from radiation necrosis on multiresolution radiomic features, reaching AUC 0.892 ± 0.055 on NIH/NIBIB-funded data.
- Designed a copula-based survival pipeline handling dependent censoring in glioblastoma; cross-validated AUC ≈ 0.77, outperforming Cox regression.
- Implemented an uncertainty quantification framework over ensemble nnU-Nets, combining entropy and mutual-information estimates to stratify erroneous segmentations; led a team to eighth place in the AMOS 2022 challenge.

Old Dominion University - *Graduate Teaching Assistant* — *Norfolk, VA* **2019 - Present**

- Ran problem sessions and laboratories for circuits, linear systems, and signals courses; built MATLAB/Simulink and Multisim exercises tying theory to application.
- Supported ECE 481W/487 Senior Design from 2021, developing course materials and ABET outcome reports.
- Mentored two NSF-REU undergraduates through summer projects on privacy-preserving and federated medical image analysis.

University of Dhaka - *Research Assistant, Microwave & Optical Fiber Lab* — *Dhaka, BD* **2014 - 2016**

- Built the lab's simulation, fabrication, and measurement capability for microstrip patch antenna research.

Selected Publications

- **Sadique, M.S.**, et al. Brain Tumor Recurrence vs. Radiation Necrosis Classification and Patient Survivability Prediction. *IEEE Journal of Biomedical and Health Informatics*, 2024.
- Temtam, A., …**Sadique, M.S.**, et al. Functional Brain Network Identification in Opioid Use Disorder Using Machine Learning of Resting-State fMRI. *Computers in Biology and Medicine*, 2025.
- **Sadique, M.S.**, et al. Class Activation Mapping and Uncertainty Estimation in Multi-Organ Segmentation. *SPIE Medical Imaging*, 2023.
- Twelve further peer-reviewed papers across MICCAI Brainlesion, SPIE Medical Imaging, and Springer book chapters.

Selected Projects

- **Adult-to-Pediatric Domain Adaptation** (2026) - gradient-reversal domain alignment with multiresolution fractal features, transferring 1,251 adult glioma cases to 228 pediatric cases; presented at SPIE Optics + Photonics.
- **Learnability Under Sampling Noise** (2025-) - Resolvable Expressive Capacity framework with eigentask decomposition, quantifying which tumor growth directions a PDE model recovers from noisy imaging.

Technical Skills

- **Languages & tools:** Python, MATLAB, R, C, Shell, Git, LaTeX, Docker, CUDA, Slurm (HPC)
- **Machine learning:** PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV; CNN, RNN, LSTM, U-Net, nnU-Net, Neural ODEs, Transformers, Diffusion Models, Transfer Learning, Domain Adaptation
- **Medical imaging:** FSL, FreeSurfer, 3D Slicer, ANTs, SimpleITK, NiBabel, ITK-SNAP, MIPAV, MedPerf
- **Statistics:** Survival analysis, copula modeling, uncertainty quantification, feature selection, PCA, LDA

Education

Ph.D., Electrical & Computer Engineering - Old Dominion University — 2018 - July 2027 (expected)
M.Sc., Electrical & Electronic Engineering - University of Dhaka — 2014 - 2016
B.Sc., Electrical & Electronic Engineering - University of Dhaka — 2010 - 2014

Honors

SPIE Medical Imaging Student Conference Support (2022, 2023) · Team Lead, Eighth Place, AMOS Multi-Organ Segmentation Challenge (2022) · BraTS Challenge participant (2021-2026) · NSICT Fellowship for M.S. Thesis, Bangladesh (2015) · IEEE Graduate Student Member; SPIE Student Member